Chapter 5: Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. Instead of asking a language model to answer only from its internal knowledge, a RAG system first searches a knowledge base for relevant information. The retrieved passages are then provided to a language model as context for generating an answer.

A basic RAG pipeline contains document ingestion, text extraction, chunking, embedding generation, vector storage, retrieval, and answer generation. Documents can be split into smaller chunks and converted into embeddings. A vector database such as ChromaDB can store these embeddings. When a user asks a question, the system embeds the question, retrieves similar chunks, and sends those chunks to an LLM such as Grok. Good chunking, relevant retrieval, and clear prompts help improve answer quality.